Lecture 5: Recursion and Divide-and-Conquer

Heading: Recursion Fundamentals
A recursive function is a function that calls itself to solve smaller instances of the same problem. Every valid recursive algorithm requires at least one base case, which terminates recursion without further recursive calls, and a recursive step that moves closer to the base case.

When a function invokes itself, a new stack frame is pushed onto the system call stack. If the recursion depth is too large without hitting a base case, stack overflow occurs. Recursion is foundational for divide-and-conquer strategies and tree traversals.

Lecture 5: Recursive Call Trees

Heading: Call Trees
We can model recursive execution as a tree of subproblems. For example, Fibonacci numbers or merge sort create recursive call trees where parent calls branch into child subproblem calls. Understanding recursive call trees is vital for tree algorithms.